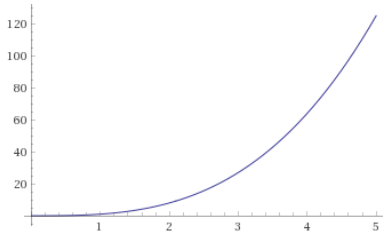
적 분 05

연습문제 풀이

연습문제 5.2

1. 정리를 이용하여 다음 함수의 그래프와 x축 사이의 영역에 대한 상계와 하계를 구하여라.

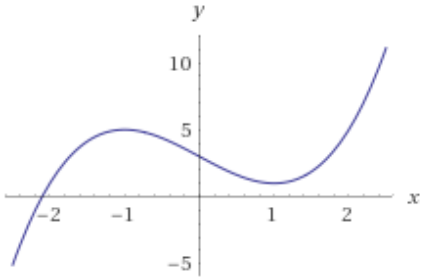
(2) $f(x)=x^3$, $[0,4]$



답: 상계 $4 \cdot 64=256$ $\therefore 256$

하계 $4 \cdot 0=0$ $\therefore 0$

(5) $f(x)=x^3-3x+3$, $[0,2]$



답: 상계 $2 \cdot 5=10$ $\therefore 10$

하계 $2 \cdot 1=2$ $\therefore 2$

연습문제 5.3

2. 다음 적분을 구하여라.

(1) $\int_0^1 4dx=[4x]_0^1=4-0$

답: 4

(2) $\int_0^2 u^{\frac{1}{2}}du=\frac{2}{3}[u^{\frac{3}{2}}]_0^2=\frac{2}{3}(2\sqrt{2}-0)$

답: $\frac{4\sqrt{2}}{3}$

연습문제 5.4

1. 다음 부정적분을 구하여라.

(1) $\int (2x^2-7x^3+4x^4)dx = \frac{2}{3}x^3-\frac{7}{4}x^4+\frac{4}{5}x^5+C$

(2) $\int \left(x^5-\frac{1}{x^4}\right)dx = \frac{1}{6}x^6+\frac{1}{3}(x)^{-3}+C$

$= \frac{1}{6}x^6+\frac{1}{3x^3}+C$

(3) $\int (x^2-\sec^2 x)dx = \frac{1}{3}x^3+\tan x+c$

(4) $\int \left(x+\frac{1}{x}\right)^2 dx = \int (x^2+2+x^{-2})dx = \frac{1}{3}x^3+2x-\frac{1}{x}+C$

(5) $\int \left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)dx = \frac{2}{3}x^{\frac{3}{2}}+2x^{\frac{1}{2}}+C$

$= \frac{2}{3}x\sqrt{x}+2\sqrt{x}+C$

연습문제 5.5

1. 주어진 치환을 이용하여 적분하여라.

(1) $\int \sqrt{4x-5}\,dx,\ \ u=4x-5$

$u=4x-5$

$\Rightarrow du=4dx$

$\Rightarrow dx=\frac{1}{4}du$

$$\int \sqrt{4x-5}\,dx=\int \sqrt{u}\,\frac{1}{4}\,du=\frac{1}{4}\left(\frac{2}{3}u^{\frac{3}{2}}\right)+c$$

$$=\frac{1}{4}\cdot\frac{2}{3}(4x-5)\sqrt{4x-5}+c$$

$$=\frac{1}{6}(4x-5)\sqrt{4x-5}+c$$

(2) $\int \cos\pi x\,dx,\quad u=\pi x$

$u=\pi x$

$\Rightarrow du=\pi dx$

$\Rightarrow dx=\frac{1}{\pi}du$

$$\int \cos\pi x\,dx=\int (\cos u)\frac{1}{\pi}\,du$$

$$=\frac{1}{\pi}\sin u+c=\frac{1}{\pi}\sin(\pi x)+c$$

(3) $\int x\cos(x^2)\,dx,\quad u=x^2$

$u=x^2$

$\Rightarrow du=2x\,dx$

$\Rightarrow dx=\frac{1}{2}du$

$$\int x\cos(x^2)\,dx=\int (\cos u)\frac{1}{2}\,du=\frac{1}{2}\int \cos u\,du$$

$$=\frac{1}{2}\sin u+c=\frac{1}{2}\sin(x^2)+c$$

(4) $\int x\sqrt{x-1}\,dx,\quad u=x-1$

$u=x-1$

$\Rightarrow x=u+1,\ du=(1)dx$

$$\int x\sqrt{x-1}\,dx=\int (u+1)u^{\frac{1}{2}}\,du$$

$$=\int \left(u^{\frac{3}{2}}+u\frac{1}{2}\right)du=\frac{2}{5}u^{\frac{5}{2}}+\frac{2}{3}u^{\frac{3}{2}}+c$$

$$=\frac{2}{5}(x-1)^{\frac{5}{2}}+\frac{2}{3}(x-1)^{\frac{3}{2}}+c$$

(5) $\int x^2\sqrt{x+3}\,dx,\quad u=x+3$

$u=x+3$

$\Rightarrow x=u-3,\ du=(1)dx$

$$\int x^2\sqrt{x+3}\,dx=\int (u-3)^2u^{\frac{1}{2}}\,du$$

$$=\int (u^2-6u+9)u^{\frac{1}{2}}\,du$$

$$=\int \left(u^{\frac{5}{2}}-6u^{\frac{3}{2}}+9u^{\frac{1}{2}}\right)du$$

$$=\frac{2}{7}u^{\frac{7}{2}}-6\cdot\frac{2}{5}u^{\frac{5}{2}}+9\cdot\frac{2}{3}u^{\frac{3}{2}}+c$$

$$=\frac{2}{7}(x+3)^{\frac{7}{2}}-\frac{12}{5}(x+3)^{\frac{5}{2}}+6(x+3)^{\frac{3}{2}}+c$$

(7) $\int 3\sin(-2x)\,dx,\quad u=-2x$

$u=-2x$

$\Rightarrow du=(-2)dx$

$\Rightarrow dx=\left(-\frac{1}{2}\right)du$

$$\int (1-5x^2)^{\frac{2}{3}}(10x)dx=\int 3\sin(u)\left(-\frac{1}{2}\right)du$$

$$=-\frac{3}{2}\int \sin(u)\,du=\frac{3}{2}\cos(u)+c$$

$$=\frac{3}{2}\cos(-2x)+c$$

2. 다음 적분을 구하여라.

(1) $\int x^3(2-5x^4)^7 dx$

$u=2-5x^4$

$\Rightarrow du=-(20x^3)dx$

$\Rightarrow (x^3)dx=-\frac{1}{20}du$

$\int x^3(2-5x^4)^7 dx=\int (u)^7(-\frac{1}{20})du$

$=-\frac{1}{20}\int u^7 du=-\frac{1}{20}(\frac{1}{8}u^8)+c$

$=-\frac{1}{160}(2-5x^4)^8+c$

(2) $\int (1+\frac{1}{x^2})(x-\frac{1}{x})^{-3}dx$

$u=x-\frac{1}{x}$

$\Rightarrow du=(1+\frac{1}{x^2})dx$

$\int (1+\frac{1}{x^2})(x-\frac{1}{x})^{-3}dx=\int u^{-3}du$

$=-\frac{1}{2}u^{-2}+c=-\frac{1}{2}(x-\frac{1}{x})^{-2}+c$

(3) $\int \sqrt{\sin 2x}\cos 2x\, dx$

$u=\sin 2x$

$\Rightarrow du=2(\cos 2x)dx$

$\Rightarrow (\cos 2x)dx=\frac{1}{2}du$

$\int \sqrt{\sin 2x}\cos 2x\, dx=\int \sqrt{u}(\frac{1}{2})du$

$=\frac{1}{2}(\frac{2}{3}u^{\frac{3}{2}})+c=\frac{1}{3}(\sin 2x)^{\frac{3}{2}}+c$

(4) $\int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos^2 x}dx$

$u=\cos x$

$\Rightarrow du=-\sin dx$

$\Rightarrow \sin dx=(-1)du$

$x=0\rightarrow u=1$

$x=\frac{\pi}{4}\rightarrow u=\frac{\sqrt{2}}{2}$

$\int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos^2 x}dx=\int_1^{\frac{\sqrt{2}}{2}} \frac{1}{u^2}(-1)du$

$=-\int_{\frac{\sqrt{2}}{2}}^{1}(u^{-2})du=\left[-\frac{1}{u}\right]_{\frac{\sqrt{2}}{2}}^{1}$

$=-1-(-\sqrt{2})=\sqrt{2}-1$

연습문제 5.6

1. 다음 적분을 하여라.

(1) $\int_{\frac{1}{9}}^{\frac{1}{4}} \frac{-1}{3x} dx$

$$\int_{\frac{1}{9}}^{\frac{1}{4}} \frac{-1}{3x} dx = -\frac{1}{3}\int_{\frac{1}{9}}^{\frac{1}{4}} \left(\frac{1}{x}\right) = -\frac{1}{3}\left[\ln|x|\right]_{\frac{1}{9}}^{\frac{1}{4}} = -\frac{1}{3}\left[\ln\frac{1}{4} - \ln\frac{1}{9}\right] = -\frac{1}{3}\left(\ln\frac{\frac{1}{4}}{\frac{1}{9}}\right)$$

$$= -\frac{1}{3}\left(\ln\frac{9}{4}\right) = -\frac{1}{3}(\ln 3^2 - \ln 2^2) = -\frac{2}{3}(\ln 3 - \ln 2)$$

$$\therefore -\frac{2}{3}(\ln 3 - \ln 2) \text{ or } \frac{2}{3}(\ln 2 - \ln 3)$$

(2) $\int \frac{x^4}{x^5+1} dx$

$u = x^5 + 1$

$\Rightarrow du = 5x^4\, dx$

$\Rightarrow x^4 dx = \frac{1}{5} du$

$$\int \frac{x^4}{x^5+1} dx = \int \frac{1}{u}\left(\frac{1}{5}\right) du = \frac{1}{5}\ln|u| + c$$

$$= \frac{1}{5}\ln|x^5+1| + c$$

(3) $\int \cot x\, dx$

$u = \sin x$

$\Rightarrow du = \cos x\, dx$

$$\int \cot x dx = \int \frac{1}{\tan x} dx = \int \frac{\cos x}{\sin x} dx = \int \frac{1}{u} du$$

$$= \ln|u| + c = \ln|\sin x| + c$$

(4) $\int \csc x\, dx$

(예제 5.24 참조)

$u = \csc x + \cot x$

$\Rightarrow du = -(\csc x \cot x + \csc^2 x) dx$

$\Rightarrow (\csc x \cot x + csx^2 x) dx = (-1) du$

$$\int \csc dx = \int \frac{\csc x(\csc x + \cot x)}{\csc x + \cot x} dx$$

$$= -\int \frac{1}{u} du = -\ln|u| + c$$

$$= -\ln|\csc x + \cot x| + c$$

$$(\csc x)' = \left(\frac{1}{\sin x}\right)' = \frac{0 \cdot sinx - 1\cos x}{[\sin x]^2} = \frac{-\cos x}{\sin x} \cdot \frac{1}{\sin x} = -\csc x \cot x$$

$$(\cot x)' = \left(\frac{1}{\tan x}\right)' = \frac{-\sin(x) \cdot sin - \cos x \cdot cosx}{(\sin x)^2}$$

$$= \frac{-(\cos^2 x + \sin^2 x)}{\sin^2 x} = -\csc^2 x$$

$$\therefore (\csc x + \cot x)' = (\csc x)' + (\cot x)' = -(\csc x \cot x + \csc^2 x)$$

(5) $\int \frac{\ln x}{x} dx$

$u = \ln x$

$\Rightarrow du = \frac{1}{x} dx$

$$\int \frac{\ln x}{x} dx = \int u\, du = \frac{1}{2} u^2 + c = \frac{1}{2} (\ln x)^2 + c$$

(7) $\int \frac{\cos x}{1 - \sin x} dx$

$u = 1 - \sin x$

$\Rightarrow du = -\cos x\, dx$

$\Rightarrow \cos x\, dx = (-1) du$

$$\int \frac{\cos x}{1 - \sin x} dx = \int \frac{1}{u} (-1) du.$$

$$= -\ln|u| + c = -\ln|1 - \sin x| + c$$